CS383 · SOFTWARE ENGINEERING

Lecture 5 Process Quality

Midterm summary

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01 Process and product quality

Product quality depends on how the product is developed. A good process usually produces a good product.

The four factors of product quality

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Process quality
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Development technology
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People quality
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Cost, time and schedule
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Project size	What determines quality
Large projects	The **Development process**.
Small projects	The developers' skills (**Developer skills**) and the technologies used.

Whatever the project size, an unrealistic schedule produces a weak product.

Appeared in the midterm

MCQ question: what does product quality depend on? The corrected answer was **Development process**.

02 Process attributes

These are attributes we use to evaluate the process itself. Each attribute answers one

question:

Attribute	The question it answers
Understandability	Is the process definition easy to understand?
Standardization	Is it based on general standards?
Visibility	Are the results of the activities and the progress clear from the outside?
Measurability	Does it make it easy to collect data so we can measure it?
Supportability	Is it supported by tools?
Acceptability	Do the engineers accept it, and can they use it?
Reliability	Does it prevent errors in the product?
Robustness	Does it keep going even when unexpected problems appear?
Maintainability	Does it evolve as the requirements change?
Rapidity	How fast is the product delivered from the specification?

Appeared in the midterm

MCQ question: which attribute focuses on how easy the process definition is to understand? The answer is **Understandability**.

03 Process improvement

Process improvement is a cycle of three stages that repeats:

#	Stage	What happens
1	**Measure**	We measure the attributes of the process according to the organization's goals. The measurement is the baseline we compare against.
2	**Analyze**	We assess the process and identify the weaknesses and the bottlenecks.
3	**Change**	We propose changes that fix the weaknesses found in the analysis.

04 Process measurement

Measurements are numerical data about the process.

Three types of process metrics

Type	Example
The time a particular activity takes	The time to complete the modeling.
The resources a particular activity needs	The effort in person-days to gather the requirements.
The number of times a particular event occurs	The number of errors found in a code review.

Watch out

Measurements are used for analysis only. The improvement itself must be driven by the objectives (**Objectives**).

Control metric and Predictor metric

Type	Its job	Example
Control metric	Supports the management of the process.	Average effort, time to repair defects.
Predictor metric	Predicts the characteristics of the software.	The number of attributes in a class, the average length of identifiers.

Goal-Question-Metric (GQM)

A method that helps you decide what to measure. It was proposed by Basili and Rombach.

Part	Meaning	Example
Goal	What the organization wants to achieve.	Shorten the product delivery time.
Question	Questions that are linked to the goal and need an answer.	How can we reduce the time to build the models?
Metric	The measurements that answer the questions.	How long does the modeling take with the current tool?

05 Process analysis and Process change

Process analysis

We study the process to understand its characteristics and how people carry it out. The aspects of the analysis:

Aspect	Example question
Adoption and standardization	Is the process documented and followed across the whole organization?
Software engineering practice	Are any good practices missing?
Organization constraints	What constraints affect the process?
Communication	Do communication problems delay the project?
Learning	How does a new employee learn?
Tool support	Which parts are supported by tools and which are not?

Process change

Modifying the current process. It can be: adding new practices, changing the order of the activities, adding or removing deliverables, or adding new roles.

There are two main challenges:

Challenge	Meaning
Resistance to change	Some people resist the change, and they may interpret the data to show that it did not work.
Change persistence	The change is dropped after a while and people go back to the old way, especially if the person who proposed it leaves the company.

06 Process classification

Type	Meaning
Informal	There is no defined model. The team chooses its own way of working.
Managed	There is a defined model that guides the development.
Methodical	The process is supported by a specific development method, such as RUP.
Improving	The process is understood in numbers, and it is analyzed and changed to improve performance.

07 Software standards

Standards define the attributes of the product or the process, and they play an important role in quality management.

Their levels

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International → National → Organizational → Project
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Each level is built on the one before it.

Why are they important?

- They collect the best practices (**Best practices**) so that we do not repeat past mistakes.
- They define what quality means in a particular setting.
- They provide continuity (**Continuity**): any new employee can easily understand the organization's standards.

Two types

Type	Applies to	Example
Product standards	The software being built.	Documentation standards (IEEE), coding standards.
Process standards	The way of working.	Defining the specification, design, and validation processes.

Problems with standards

- Engineers may see them as overprescriptive or out of date.
- They involve a lot of form filling (bureaucratic).
- Some are not supported by tools, so applying them by hand is hard and takes time.

ISO 9001

A set of international standards for building quality systems, including for software. It sets out the quality principles and describes the quality processes.

08 Common mistakes

Mistake	Correct
Confusing Visibility with Understandability	**Visibility** is how clear the progress is from the outside. **Understandability** is how easy the definition is to understand.
Swapping Control and Predictor	**Control** is for managing the process. **Predictor** is for predicting the characteristics of the software.
Saying that measurements drive the improvement	Measurements are for analysis. The improvement is driven by the objectives.
Reversing the order of GQM	Goal, then Question, then Metric.
Saying that the quality of large projects depends on the developer's skill	It depends on the **Development process**.

09 Test yourself

1. Which attribute asks: does the process keep going when unexpected problems appear?
2. Which attribute asks: is the progress clear from the outside?
3. "The number of attributes in a class" is an example of which type of metric?
4. What are the stages of the process improvement cycle, in order?
5. What are the two challenges in process change?

Answers

#	Answer
1	**Robustness**.
2	**Visibility**.
3	**Predictor metric**.
4	**Measure**, then **Analyze**, then **Change**.
5	**Resistance to change** and **Change persistence**.